The Sound of Code-Switching: Prosodic Signatures of Spanish-English Speech

Author Anonymous[1], *Co-author Anonymous*[2], *Final author Anonymous*[1]

[1]Author Affiliation
[2]Co-author Affiliation

author@university.edu, coauthor@company.com, author@university.edu

Abstract

Code-switching (CSW) has been studied in many linguistic frameworks, but little is known about the *prosody* of such multilingual speech. This has left open the questions: Is CSW prosodically different from monolingual productions in spontaneous conversation? And how are variations influenced by speaker proficiency and/or specifically multilingual features of speech? To answer these, we examine a large-scale Spanish-English corpus over a host of language-independent prosodic features, finding that CSW prosody differs meaningfully from monolingual prosody in both English and Spanish, with greater influence on this variation coming from speaker-specific factors. Our work is the first to apply such a language-generalizable approach to studying spontaneous CSW prosody, and provides novel insights on multilinguality that have important implications for the recognition and synthesis of code-switched speech.

Index Terms: code-switching, multilingual, prosodic analysis

1. Motivation

Speakers who *code-switch* alternate between multiple language varieties during conversation [1]. Code-switching (CSW) is performed between or within utterances and is common among the growing majority of global multilingual speakers, particularly in spontaneous, informal speech. Spoken CSW has been studied through various computational frameworks, spanning morpho-syntactic analyses [2, 3], socio- and paralinguistic investigations [4, 5], and discourse-functional studies [6, 7]. In contrast, relatively little is known about the *prosodic* nature of code-switched speech, leaving great scope for exploring this uniquely spoken dimension of CSW. While recent work has begun to study prosody in multilingual domains [8], our understanding of how CSW compares prosodically to monolingual speech in any of the involved languages remains incomplete.

Prior work on this topic has taken a *perception*-oriented approach, proposing that phonetic cues such as pitch accent, speaking rate, and voice onset time (VOT) allow listeners to *anticipate* switches in Spanish- and Mandarin-English [9, 10, 11]. In some cases, however, it is unclear whether these cues are the same as those actually produced differently by bilingual speakers. To address this, other studies have focused on prosodic patterns in speech *production*, showing that code-switches tend to occur across, rather than within, intonational units in Spanish-English [12, 13, 14]. But, these have provided limited insight into the prosodic characteristics of CSW itself.

Among studies specifically investigating the nature of CSW prosody, claims range from CSW being no different from monolingual prosody in French- and Spanish-English [14, 15], to CSW as fundamentally distinct from monolingual speech in terms of prosodic-syntactic factors [13]. Intermediate claims suggest *prosodic blending* in CSW; evidence includes the presence of intonational contours from both languages in Spanish-English utterances [9], changes to consonantal VOT expression near CSW boundaries [10], and differences in Urum-Russian and Korean-English vowel quality relative to monolingual speech [16, 17]. Further work has studied hyperarticulation of French- and Spanish-English vowels during CSW [18, 19] and VOT in Greek-English [20], showing that these effects can vary by social context and speaker language dominance [20, 21]. However, the role of language proficiency is less conclusive among balanced bilinguals with roughly equal proficiency across languages [22, 23].

Despite the breadth of related work, it is difficult to generalize about the nature of CSW prosody; all of the above studies, except for [13], examined only *read* speech produced by a handful of speakers, with most analyses done either over single lexical items or phonemes in isolation, sometimes within specific syntactic constraints, or over a few features from a narrow prosodic class. This has resulted in insightful but contradictory and overly fine-grained analyses focused on elicited speech.

We propose to fill the gap in our understanding of *naturally-occurring* CSW prosody and improve upon the limitations of prior work by studying the prosodic production of Spanish-English bilingual speakers in a large-scale corpus of CSW. We examine the extent of variation between code-switched and monolingual prosodic profiles across both represented languages and whether any differences can be explained by speaker demographic attributes or multilingual linguistic characteristics. Specifically, we ask **RQ1:** Is there utterance-level variation across a suite of language-independent pitch, energy, and duration features between code-switched and monolingual spontaneous speech? And, **RQ2:** How is this variation influenced by **(a)** speaker proficiency and **(b)** linguistic characteristics of multilingual speech? To address these **RQs**, we build on [8] and find that CSW prosody differs meaningfully from monolingual prosody in both Spanish and English, with speaker-specific factors exhibiting stronger influence than multilingual linguistic attributes. Overall, we show how and explain why spontaneous CSW is prosodically *distinct* from monolingual speech. Our main contribution is a clearly interpretable, novel insight into the prosodic nuances of CSW, which has important implications for naturalistic multilingual speech synthesis and recognition.

2. Method

Corpus. We examine the Bangor Miami corpus (BM) of spontaneous, informal Spanish-English speech [24]. BM consists of 35 hours of recorded dialogue across 56 conversations, corresponding to over 46,900 transcribed utterances exchanged between 84 speakers (52 female, 32 male). The dataset comprises a mix of monolingual English, monolingual Spanish, and code-switched Spanish-English utterances, all manually annotated with token-level language identification (LID) tags. The corpus also includes demographic information on each speaker,

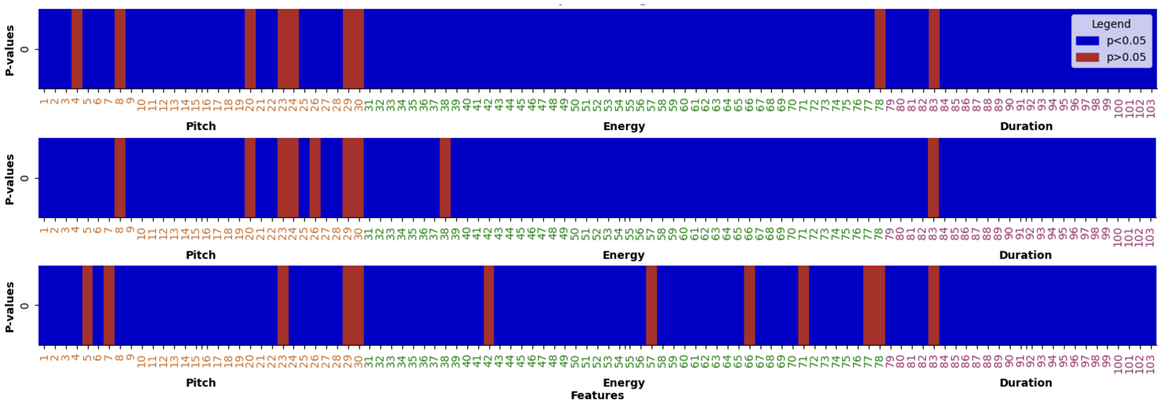


Figure 1: *p-values for prosodic feature distribution comparisons between code-switched and combined monolingual English & Spanish (top), monolingual English (middle), and monolingual Spanish (bottom) utterances. Blue indicates statistical significance; red indicates insignificance. Associated Cohen's* d *values generally fall within* $0.2 < d < 0.8$ *(i.e. small to medium effects) for significant features.*

covering their medium of schooling, years of language experience, self-reported ability in each language, age, and each of their parents' primary language. We treat these as indicators of speaker language proficiency, as done in [25].

Feature extraction. We extract and vectorize the DisVoice[1] set of 103 utterance-level prosodic features derived from three categories – fundamental frequency (F0), energy, and duration – and six functionals across voiced and unvoiced segments. For consistency with prior work [8], we extract features from denoised BM data [26]. We also quantify utterance-level CSW quantity and frequency using M- and I-indices [27, 28], and use prior annotations [26] of CSW direction (*en* $\rightarrow$ *es* vs. *es* $\rightarrow$ *en*) and strategy (*insertional* vs. *alternational* [29]).

Statistical testing. Throughout the work, we compare the prosodic feature distributions of monolingual English and Spanish to those of code-switched Spanish-English over multiple subsets of the data using feature-level independent t-tests with the Benjamini-Hochberg correction method, and support these with Cohen's d effect sizes and z-tests of proportions.

Modeling. We complement our statistical results with relevant 1) unsupervised and 2) supervised model analyses. For 1), we use a `scikit-learn 1.6.1` k-means clustering model with default hyperparameters. For 2), we follow [8] by using LID as a tool for understanding prosody; we evaluate the binary classification performance of a pre-trained prosodic-LID model: `Whisper-base` [30]. This model was pre-trained for monolingual automatic speech recognition (ASR) in 97 languages, including English and Spanish, but allows access to intermediate LID tokens, which can be restricted to a subset of languages to enforce binary evaluation. We study off-the-shelf and fine-tuned performance of this model, using a 90/10 fine-tuning/inference split of speaker-level code-switched BM data.

3. Results

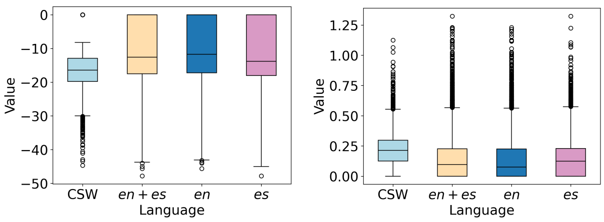
3.1. CSW differs prosodically from monolingual speech.

We first explore whether CSW prosody differs from monolingual prosody. We find that the bulk of prosodic features in code-switched utterances are significantly different ($p < 0.05$) from the same ones in monolingual English and Spanish, both considered together and separately (Figure 1). Since speakers in BM are all adults from Miami in the U.S. state of Florida, who converse informally on shared topics (e.g. family life, relationships, hobbies, etc.), and each produce utterances in monolingual English, monolingual Spanish, and code-switched Spanish-English, the impact of these speaker variables on the variation we find is minimal. Across comparisons, duration features drive the majority of this prosodic variation in CSW, as at least 96% of these vary significantly from their counterparts in monolingual speech. Energy features are the next most important, with no less than 87.5% of features differing significantly from corresponding monolingual speech. Pitch features contribute the least, as up to 30% of these features are similar to their monolingual analogues. These group-level patterns are the opposite of those found in [8], and help to distinguish the prosody of monolingual speech within a multilingual discourse context from that of multilingual speech itself.

We inspect these prosodic differences more closely and highlight key observations from qualitative comparisons between code-switched and monolingual speech. F0 contours of initial and final voiced segments generally have higher mean values in code-switched utterances than in monolingual ones, with greater variation in code-switched F0, particularly for initial segments. This is true in comparisons with both monolingual English and Spanish, with consistently greater Δs between CSW and Spanish than between CSW and English. In terms of energy, initial and final voiced segments have lower mean and maximum values across CSW relative to monolingual speech in both languages. This is also true for minimum energy in unvoiced segments. Additionally, across voiced and unvoiced segments of code-switched utterances, variation in energy features is greater than in monolingual utterances. Finally, the mean and standard deviation in duration of voiced and unvoiced segments, as well as pauses, are greater during CSW than in monolingual English and Spanish. The ratio of duration of pauses to combined voiced and unvoiced segments is also higher in CSW than monolingual speech. We visualize some of these trends in Figure 2. Cumulatively, we find that CSW tends to sound higher-pitched, quieter, and more disjointed than both monolingual English and Spanish. The consistent pattern of heightened *variation* across code-switched prosodic feature groups is particularly striking, and echoes prior work on the diagnostic value of feature variation over absolute values [31].

Having found some evidence of prosodic patterns that distinguish code-switched utterances from monolingual speech in both English and Spanish, we examine whether these are salient

[1] https://disvoice.readthedocs.io/en/latest/Prosody.html



(a) (b)

Figure 2: *Visualizing trends in (a) mean energy and (b) mean duration of intial voiced segments across utterance types.*

enough for a simple unsupervised model to learn and apply. Our k-means ($k = 2$) clustering model demonstrates reasonable ability to do so in multiple clustering scenarios, achieving an average accuracy of approximately 85% on differentiating code-switched and monolingual utterances based on prosody alone (Table 1). Visual inspection of clustering outputs (e.g., Figure 3) reveals a distinct boundary between the code-switched and monolingual clusters, reinforcing our findings so far. Overall, our statistical, qualitative, and unsupervised modeling results provide complementary evidence of meaningful prosodic differences between code-switched and monolingual speech, which together motivate the remainder of the work.

Table 1: *k-means clustering performance across comparison settings. The first comparison (*en *vs.* es*) serves as a baseline.*

Cluster comparison	**Accuracy**
Monolingual *en* vs. Monolingual *es*	0.636
All monolingual vs. code-switched	0.843
Monolingual *en* vs. code-switched	0.837
Monolingual *es* vs. code-switched	0.868

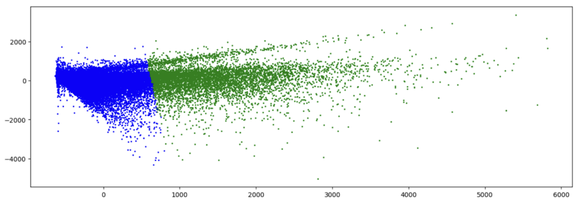


Figure 3: *Clustering all monolingual (blue) vs. code-switched (green) utterances. Axes represent the relevance of selected prosodic feature groups post-PCA; the highest loadings for PC1 (x-axis) and PC2 (y-axis) correspond to greater mean , max., and SD in energy and pitch features. Visualizations for clustering English vs. CSW and Spanish vs. CSW are almost identical.*

3.2. Prosodic variation in CSW is shaped by speaker-level language proficiency characteristics.

Having found that CSW differs prosodically from monolingual speech, we now must explain these differences. Since prior work disagrees on the influence of speaker proficiency on prosody [20, 22], we examine proficiency indicators relative to group-level prosodic features. First, we identify which language proficiency factors correspond to the greatest prosodic differences across demographic groups. We then apply our findings towards examining whether CSW prosody more closely resembles the monolingual prosody of the higher-proficiency language, as indicated by the most salient demographic factors.

To begin, we construct two comparison groups for each language proficiency feature: for categorical features, we bin speakers according to their indicated language (*es* or *en*); for continuous ones, we assign speakers to groups based on their reported ratio value ($>$ 1: *es*; $<$ 1: *en*). Then, across each pair of comparison groups corresponding to a given proficiency feature, we compare prosodic feature distributions over each of three utterance types: 1) CSW, 2) monolingual English, and 3) monolingual Spanish. In other words, in each of the three cases, we compare prosodic productions of the *same* type of utterance between speakers with opposite proficiency attributes.

We find that language proficiency factors generally correspond to prosodic differences in monolingual speech across speaker groups, but do not show consistent effects in CSW. For example, over 80% of prosodic features are significantly different across Spanish utterances produced by speakers whose secondary schooling was in English relative to those whose secondary schooling was in Spanish. The extent of differences increases to 85% over English utterances for the same comparison. Similarly, over 72% and 77% of prosodic features are significantly different across monolingual English and Spanish utterances, respectively, as produced by speakers with English-medium primary schooling relative to those with Spanish-medium primary schooling. In contrast, for code-switched utterances, the only language proficiency factor for which a majority (51%) of prosodic features differ significantly between speaker groups is years of experience. In sum, language proficiency, as defined categorically by language of schooling, appears to play a greater role in shaping prosodic variation in monolingual rather than code-switched speech.

Based on these findings, we hypothesize that speakers produce CSW prosody that is *more similar* to the monolingual prosody of their *higher-proficiency* language, as indicated by primary and secondary school media, than that of their lower-proficiency language. Our experimental findings support this hypothesis. Speakers whose primary school medium was Spanish show fewer statistically significant differences between their code-switched and Spanish prosody relative to those between their code-switched and English prosody: $\Delta_{CSW/en-CSW/es} = 88\% - 72\% = 16\%$. We find a consistent pattern with respect to secondary schooling in Spanish: $\Delta_{CSW/en-CSW/es} = 89\% - 74\% = 15\%$. Similarly, considering speakers whose schooling was in English, $\Delta_{CSW/es-CSW/en} = 95\% - 86\% = 9\%$ at the primary level and $\Delta_{CSW/es-CSW/en} = 93\% - 83\% = 10\%$ at the secondary level. These Δs are all statistically significant with $p < 0.05$ per z-tests of proportions, and correspond to 15-20% of baseline significant differences between monolingual English and Spanish prosodic features in each comparison, suggesting a moderate but meaningful relationship between language proficiency and the prosodic nature of CSW. This relationship is stronger among speakers with greater Spanish proficiency, with more pronounced effects of secondary school medium, which ties directly into previous findings on CSW linguistic behavior and language proficiency in BM [25]. So, we have preliminary indications of speaker-level characteristics that may drive prosodic differences between CSW and monolingual speech.

We now define a LID task to assess the strength of this pattern: a binary classifier that can internalize the prosodic relationship between proficiency and CSW should identify a code-switched utterance by a Spanish-proficient speaker as monolingual Spanish, and one produced by an English-proficient speaker as monolingual English, given prosodic CSW inputs and only these two output classes. Thus, we use prosodic-LID as a proxy to identify which language CSW resembles *more*.

Table 2: *Comparing pre-trained (PT) and fine-tuned (FT) model accuracy & class-wise F1 score on binary prosodic-LID. Fine-tuned Whisper-base uses an additional classifier head (CH).*

Model	**Accuracy**	**F1-Score** *en*	*es*
`Whisper-base` (PT)	0.64	0.74	0.44
`Whisper-base` (FT)	0.59	0.69	0.41
`Whisper-base` (FT + CH)	0.89	0.91	0.85

The pre-trained `Whisper-base` model's off-the-shelf (baseline) performance is underwhelming on prosodic-LID for code-switched data, with particular generalization issues between language classes (Table 2). We fine-tune `Whisper-base` to improve LID performance, but find that it fails to outperform its off-the-shelf baseline, though it still surpasses a random guessing baseline. Such poor performance may be due to the model's original sequence labeling objective, rendering it inadequate for direct application to our classification task. To tackle this, we append a linear classifier head to its encoder, which leads to notable improvement in performance on the held-out test set of unseen speakers, both overall and for each language individually. This set of fine-tuned `Whisper-base` results suggests that the model may well learn how CSW prosody and language proficiency are related, reinforcing the role of speaker characteristics in shaping prosodic variation in CSW, and providing support for the salience of this relationship.

3.3. Prosodic variation in CSW is influenced by linguistic characteristics of multilingual speech.

To complete our study, we apply another lens to explain differences between code-switched and monolingual prosody: the properties of CSW itself. We specifically examine how CSW richness, direction, and strategy relate to the extent to which code-switched prosody differs from monolingual prosody. First, we check whether the prosody of utterances containing abundant or frequent code-switches differs from monolingual speech to a greater extent than that of utterances containing sparing or infrequent code-switches. We aggregate utterance-level richness metrics by speaker to construct groups of high vs. low-quantity and -frequency code-switchers, binning a speaker [above/below] the corpus-level median for each metric in the corresponding [high/low] group. We find that high-quantity code-switchers show greater or equal proportions of significant feature-level differences between their CSW and monolingual prosody compared to low-quantity code-switchers ($CSW/en\Delta_{high_M - low_M} = 91\% - 81\% = 10\%$; $CSW/es\Delta_{high_M - low_M} = 85\% - 85\% = 0\%$.) This pattern replicates for high- vs. low-frequency code-switchers ($CSW/en\Delta_{high_I - low_I} = 93\% - 84\% = 9\%$; $CSW/es\Delta_{high_I - low_I} = 84\% - 84\% = 0\%$), and is again more notable for differences between CSW and English. Both non-zero Δs correspond to z-test p-values < 0.05, and about 11% of baseline corpus-level significant differences between monolingual English and Spanish. These patterns hold when we control for speaker gender, and overall suggest that prosodic differences between code-switched and monolingual speech may be slightly enhanced with greater CSW richness.

Next, we examine whether prosodic differences relative to monolingual speech are more pronounced in *es→en* or *en→es* CSW. In comparisons to both monolingual Spanish and English, *es→en* shows slightly more variation than does *en→es* ($\Delta_{es \to en - en \to es} = 87\% - 84\% = 3\%$ and $\Delta_{es \to en - en \to es} = 91\% - 87\% = 4\%$, respectively). However, these absolute differences are not significant according to z-tests, so our results fail to support a relationship between CSW direction and prosodic variation relative to monolingual speech in this corpus.

Finally, inspired by [32], we examine whether code-switched prosodic differences depend on the CSW strategy in use. Relative to English, insertional CSW differs from monolingual speech to a similar extent to alternational CSW from the same ($\Delta_{CSW_I - CSW_A} = 91\% - 87\% = 4\%$; z-test: $p > 0.05$). In contrast, relative to Spanish, insertional CSW differs from monolingual speech more than alternational CSW does ($\Delta_{CSW_I - CSW_A} = 91\% - 81\% = 10\%$; z-test: $p < 0.05$; $\equiv 17\%$ of baseline prosodic differences between the two strategies), with relevant differences concentrated in pitch and duration features. Though slightly inconclusive, these results provide promising indications that CSW strategy may indeed play a role in shaping prosodic variation in CSW. Overall, among the properties studied, variation in CSW richness and strategy drive most of the differences that distinguish CSW and monolingual prosody; CSW direction appears not to play an influential role. These preliminary findings provide additional support for CSW differing prosodically from monolingual speech and contribute an additional explanatory dimension for this phenomenon.

4. Discussion

Most prior studies of prosody have focused on monolingual speech settings. The findings of those on CSW prosody have been difficult to generalize (see Section 1). Our work adds novelty, nuance, and clarity to the broad landscape of multilingual prosodic work through a language-generalizable approach to studying spontaneous CSW at a higher level of abstraction than any individual prosodic feature. We investigate and interpret how the prosody of code-switched Spanish-English in BM differs from monolingual speech in both languages, and demonstrate that this difference can be explained to varying degrees by speaker proficiency and linguistic characteristics specific to multilingual speech. Further, the variation we find is salient enough to be learned by unsupervised and end-to-end predictive LID models; the latter have been particularly underutilized as diagnostic tools in prosody research, but our application of them demonstrates their largely untapped value in enabling quantitative interpretation of latent variation.

We conclude that spontaneous Spanish-English CSW is prosodically unique, with slightly greater similarity to monolingual Spanish than English in this corpus. The qualitative patterns of CSW prosody that we uncover mirror those found for group-level contributions to prosodic differences in prior work [8]; we provide an additional dimension of insight across types of utterances spoken in spontaneous multilingual conversation by demonstrating how CSW prosody relates to that of monolingual speech uttered in a broadly multilingual context. Combined with [8], this enables transitive inference of how CSW compares prosodically to monolingual speech from a monolingual context, building a more complete picture of prosody in diverse speech settings. In future work, it would be valuable to examine how our results generalize to dialectal CSW, interact with interlocutor identity and entrainment, and influence real-world speech systems, e.g., by improving unnatural intonation in generated bilingual speech or facilitating language learning.

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